

SURAJ KUMAR

Bhubaneswar, Odisha

+91-8789162493

surajrajnkh1244@gmail.com

LinkedIn

GitHub

LeetCode

PROFESSIONAL SUMMARY

Computer Science undergraduate (2027 Batch) specializing in Backend and Full-Stack Development. Experienced in building scalable real-time web applications using the MERN stack, WebSockets, Docker, and AWS cloud technologies. Strong foundation in REST API design, authentication systems, database management, and Data Structures & Algorithms.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), C++, SQL

Frontend: React.js, HTML, CSS, Vite

Backend: Node.js, Express.js, REST APIs, WebSockets (Socket.IO), JWT Authentication

Databases: MongoDB, MySQL

Tools: Git, GitHub, Postman, Docker, AWS (ECS, Fargate, ALB)

Core Concepts: Data Structures, Algorithms, OOPS, API Design, Real-Time Systems, Client-Server Architecture

EDUCATION

Kalinga Institute of Industrial Technology

B.Tech in Computer Science Engineering

2023 – Present

CGPA: 8.98/10

S T Raza International School (CBSE)

Class XII

2022

83.8%

PROJECTS

Real-Time Code Collaboration Platform

2026

React.js, Node.js, Express.js, Socket.IO, Yjs, Docker, AWS ECS

- Built a collaborative code editor enabling multiple users to edit code simultaneously in real time.
- Used Socket.IO and Yjs (CRDT) for low-latency synchronization and conflict-free collaboration.
- Integrated room-based collaboration and scalable backend communication using WebSockets.
- Containerized and deployed the application on AWS ECS using Docker and load balancing.
- Resolved deployment issues including routing, caching, and CORS handling.
- **Live:** Demo **GitHub:** Repository

ForgeChat — AI Powered Chat Application

2025

React.js, Node.js, Express.js, MongoDB

- Developed a ChatGPT-style AI application with secure authentication and AI-generated responses.
- Integrated GPT API based response handling through backend REST APIs.
- Implemented JWT authentication, MongoDB schema design, and session management.
- Improved backend structure by organizing API routes and optimizing database queries.
- Deployed the application on Render with environment configuration and routing support.
- **Live:** forgechat.onrender.com **GitHub:** Repository

Talk-Space — Real-Time Messaging Platform

2025

React.js, Node.js, Express.js, MongoDB, Socket.IO

- Built a real-time messaging platform supporting instant communication between users.
- Implemented WebSocket communication using Socket.IO for low-latency message delivery.
- Developed REST APIs for user management, authentication, and message handling.
- Integrated JWT-based authentication and MongoDB for secure session and chat storage.
- Optimized backend workflows and handled deployment configuration on Render.
- **Live:** talk-space-z1i1.onrender.com **GitHub:** Repository

ACHIEVEMENTS

- Solved 490+ Data Structures and Algorithms problems on LeetCode covering topics including Graphs, DP, and Greedy.
- Achieved a LeetCode contest rating of 1570, ranking in the top ~30% globally.
- Built and deployed scalable real-time applications using Docker, AWS ECS, and WebSockets.